

Landmark CNN Architectures II

ResNet & Inception

Computer Vision
Week 4 – Lecture 8

Quick Recall: Where We Left Off

AlexNet (2012)

Deeper CNN + ReLU + GPU training.

The Big Question

"If deeper networks learn better features, why don't we just keep adding layers?"

LeNet (1998)

Early CNN for digit recognition.

VGG (2014)

Depth via repeated 3×3 convolutions.

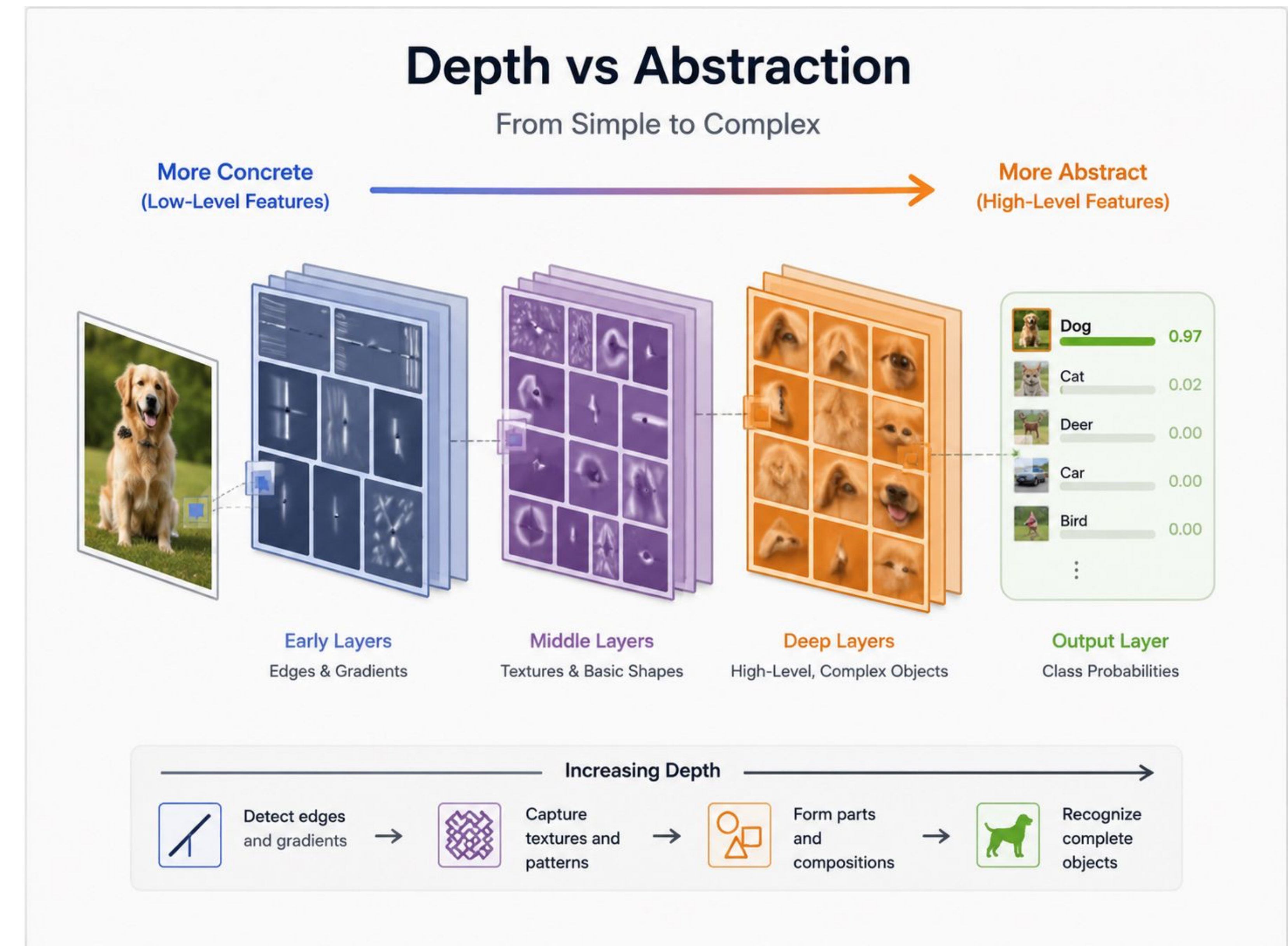
Is Deeper Always Better?

The Intuition for Depth:

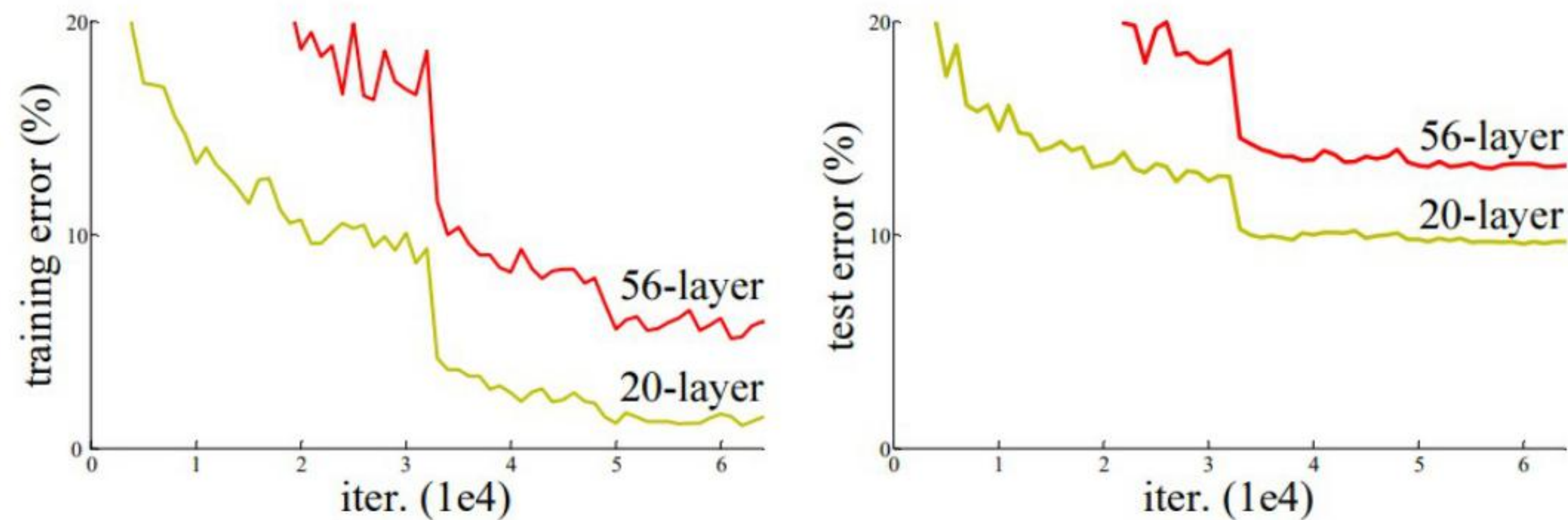
-  **Early Layers:** Learn simple edges and gradients.
-  **Middle Layers:** Learn textures and basic shapes.
-  **Deep Layers:** Learn high-level, complex objects.

The Reality Check:

-  Beyond a certain depth, adding layers can actually **increase** training error.
-  Crucially, this is **NOT overfitting**.



The Degradation Problem



- 
Observation: A 56-layer network yields higher training error (and test error) than a 20-layer network.
- 
The Paradox: A deeper model should theoretically easily mimic the shallower model by just mapping an identity function for extra layers.
- 
Conclusion: Current solvers fail to optimize these highly complex networks.
- 
"This is an optimization problem — not simply overfitting."

Why Is Training Very Deep Networks Difficult?



Vanishing Gradients

As gradients are backpropagated through dozens of layers, they multiply and shrink to near zero, stopping learning.



Information Degradation

The original image signal gets washed out or distorted after passing through too many non-linear transformations.



Identity Mapping

Traditional non-linear layers have extreme difficulty learning to just "pass the data through unchanged" (identity).

The Core Idea Behind ResNet

 Introduced by **Kaiming He et al.** (Microsoft Research, 2015).

 Winner of the ImageNet 2015 classification challenge.

 **Central Idea:** Don't force the layers to learn the complete transformation $H(x)$.

 Instead, let the layers learn a **residual function** $F(x)$, and add the input x back in!

$$H(x) = F(x) + x$$

Residual Learning: Intuition

Traditional Block

The network tries to learn an entirely new representation from scratch.

$$x \rightarrow [\text{Layers}] \rightarrow H(x)$$

Difficult to learn identity mapping.

Residual Block

The network learns "What needs to change from the input?"

$$\begin{array}{ccccc} x & \rightarrow & [\text{Layers}] & \rightarrow & F(x) \\ & \searrow & & & \downarrow \\ & & & + & \rightarrow & H(x) \end{array}$$

Identity is easy: just push $F(x)$ to zero!

Anatomy of a Residual Block

-] **Input** x splits into two paths.
- ≡ **Main Path:** Passes through Conv → Batch Norm → ReLU → Conv → Batch Norm.
- ▶▶ **Shortcut Path:** Skips the convolutional layers entirely.
- + **Element-wise Addition:** The shortcut x is added to the main path output $F(x)$.
- ✓ Final ReLU is applied *after* the addition.

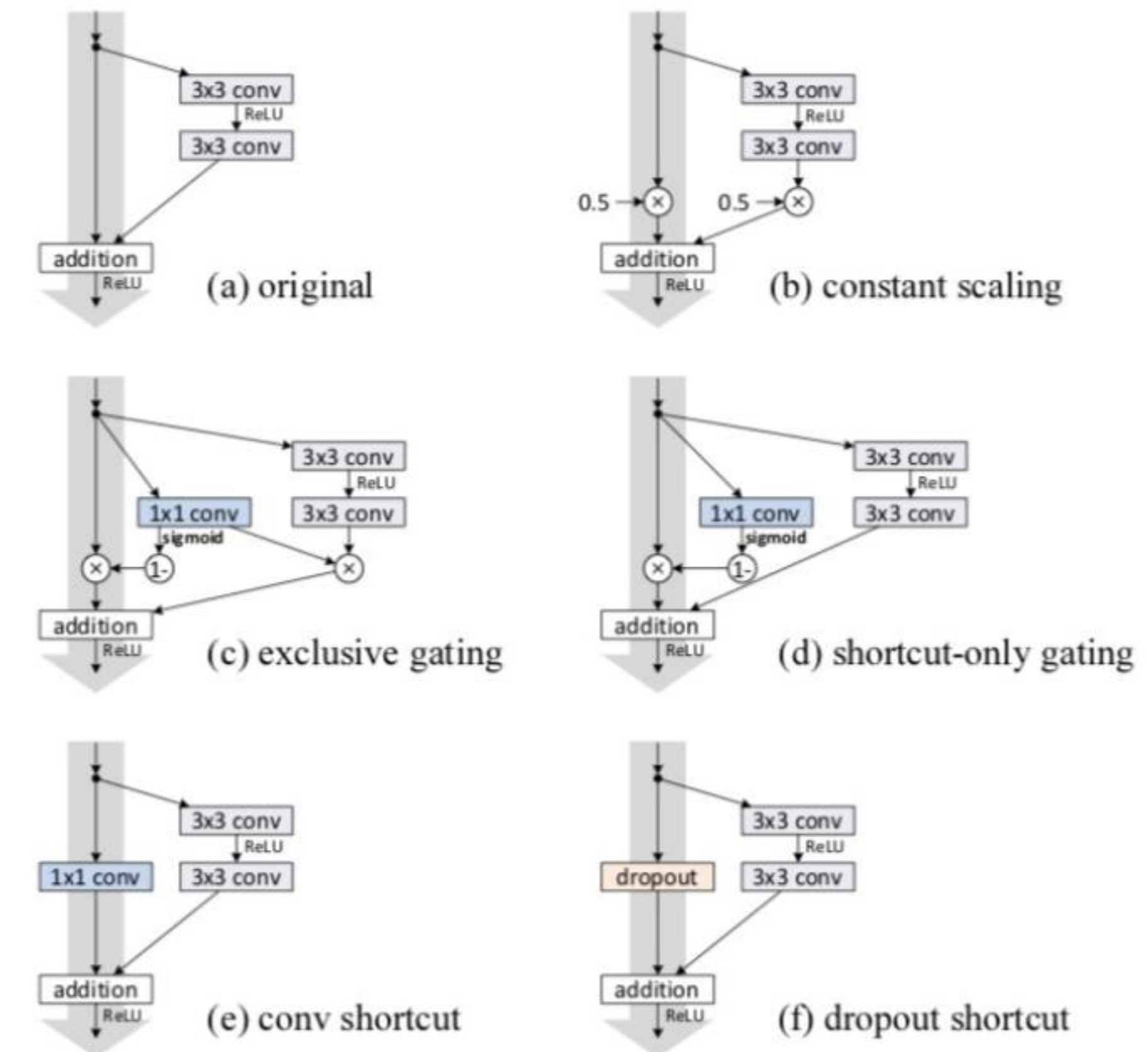


Figure 2. Various types of shortcut connections used in Table 1. The grey arrows indicate the easiest paths for the information to propagate. The shortcut connections in (b-f) are impeded by different components. For simplifying illustrations we do not display the BN layers, which are adopted right after the weight layers for all units here.

How Skip Connections Help



Easier Gradient Flow

During backpropagation, gradients flow directly back through the **skip connection**. This completely circumvents the vanishing gradient problem.



Preserves Information

The original signal is carried forward flawlessly. No matter how deep the network, the input features are always accessible.



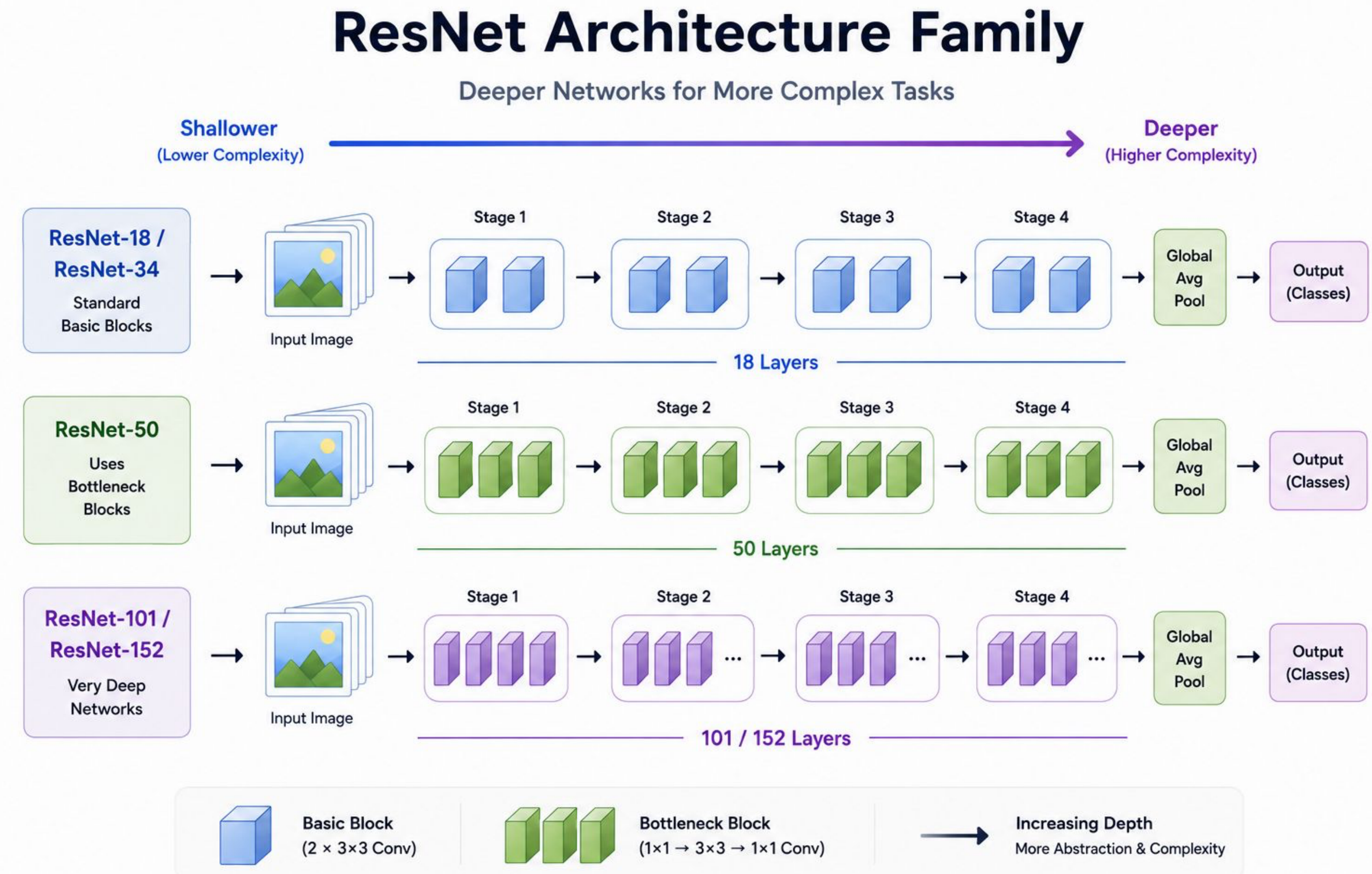
Enables Extreme Depth

Because identity is easy to learn (set weights to 0), we can stack 100+ layers. Unnecessary layers just become pass-throughs.

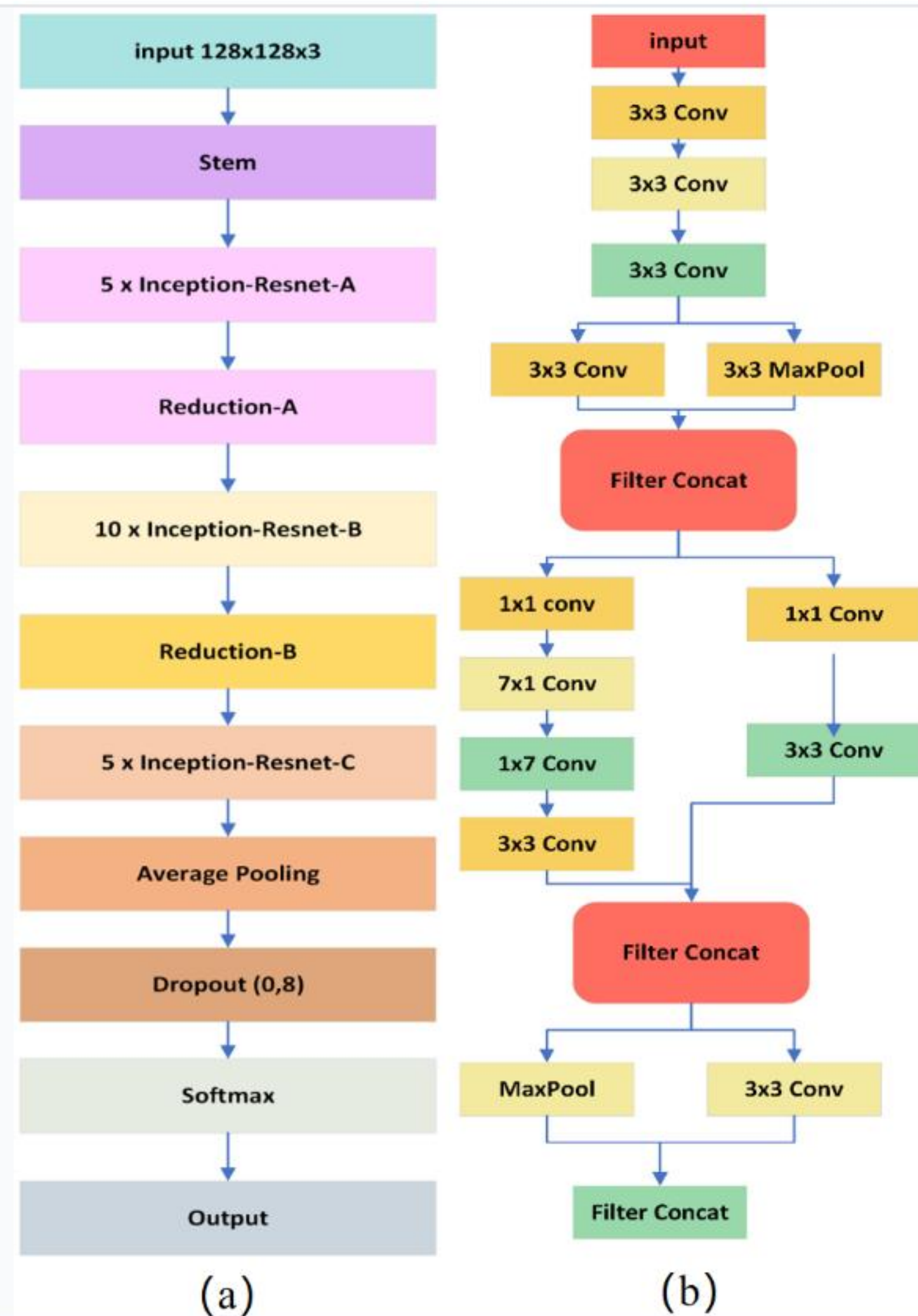
ResNet Architecture Family

ResNet is not one network, but a scalable family based on depth:

-  **ResNet-18 / ResNet-34:** Utilize standard Basic Blocks (two 3×3 convolutions).
-  **ResNet-50:** The standard workhorse. Introduces Bottleneck blocks.
-  **ResNet-101 / ResNet-152:** Extremely deep architectures for highly complex tasks, entirely reliant on Bottleneck blocks.



Bottleneck Residual Blocks



Used in ResNet-50 and deeper:

- 1×1 Conv:** Reduces the number of channels (compressing the data).
- 3×3 Conv:** Performs the heavy spatial processing on the reduced channels.
- 1×1 Conv:** Restores the channels back to their original depth for addition.
- Why?** Drastically reduces computational load while allowing the network to maintain immense depth.

Why ResNet Was a Breakthrough



Shattered Limits

Successfully trained networks with 100+ layers, completely bypassing the degradation problem that plagued VGG.



Standard Backbone

Became the default pre-trained backbone for Object Detection, Semantic Segmentation, and Medical Imaging.



Massive Influence

Skip connections are now ubiquitous. They are found in Transformers, U-Nets, and nearly all modern AI architectures.

"ResNet solves depth. But can we also make CNNs smarter about what filter size they use?"

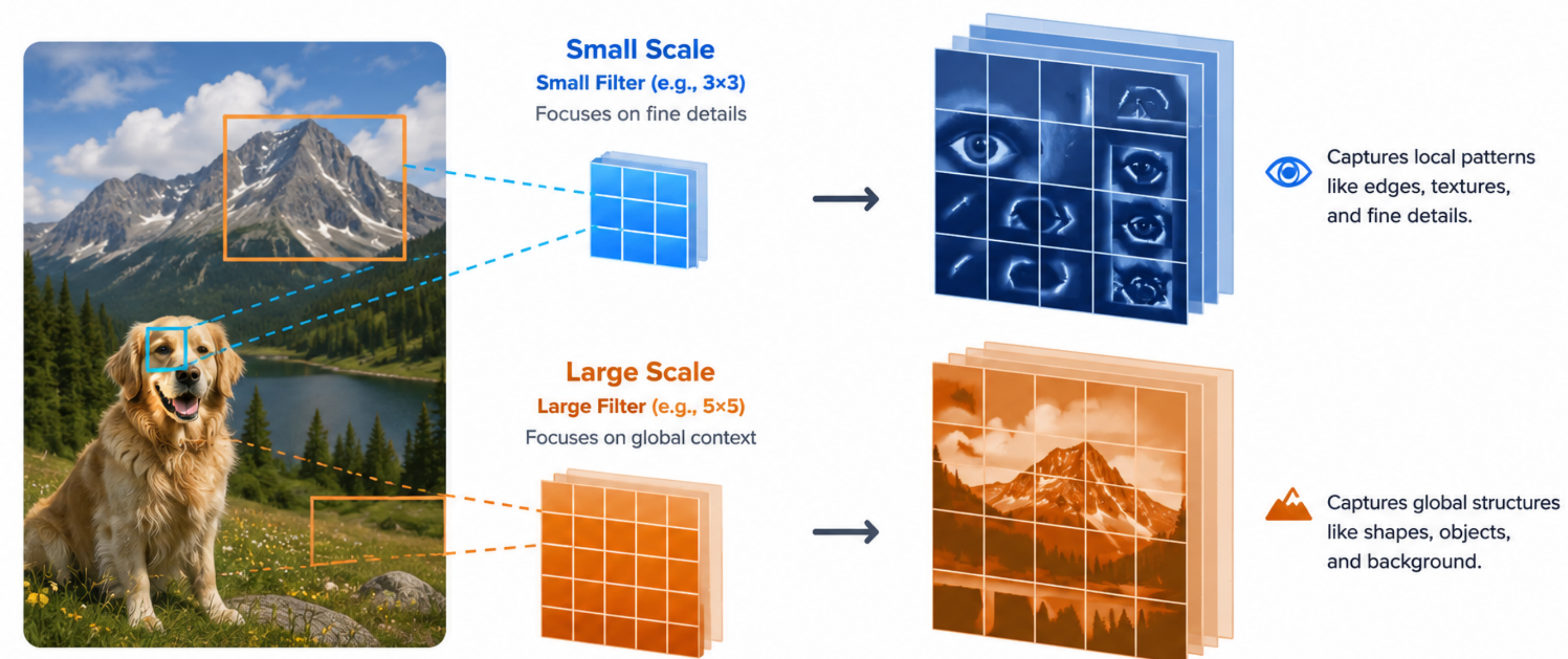
The Problem of Choosing Filter Size

When designing a CNN, how do we choose the kernel?

- ? Should we use 1×1 ? 3×3 ? 5×5 ? Or Pooling?
- 👁 Different visual patterns require different **receptive fields**.
- 🐕 **Small features:** A dog's eye needs a small filter (e.g., 3×3) to capture local detail.
- ⚠ **Large features:** A mountain background needs a large filter (e.g., 5×5) to capture global context.

Multi-Scale Features

Capturing Details at Different Scales



Why Both
Scales Matter?



Small filters capture
“what” in detail.

+



Large filters capture
“where” in the bigger
picture.

=



Together, they help the
network understand
images better.

The Inception Architecture

Introduced as GoogLeNet (Inception v1) in 2014

The Core Idea:

"Why choose one filter size when we can use several simultaneously?"

The Inception Module (Naive)

Instead of one sequential path, data flows through parallel branches:

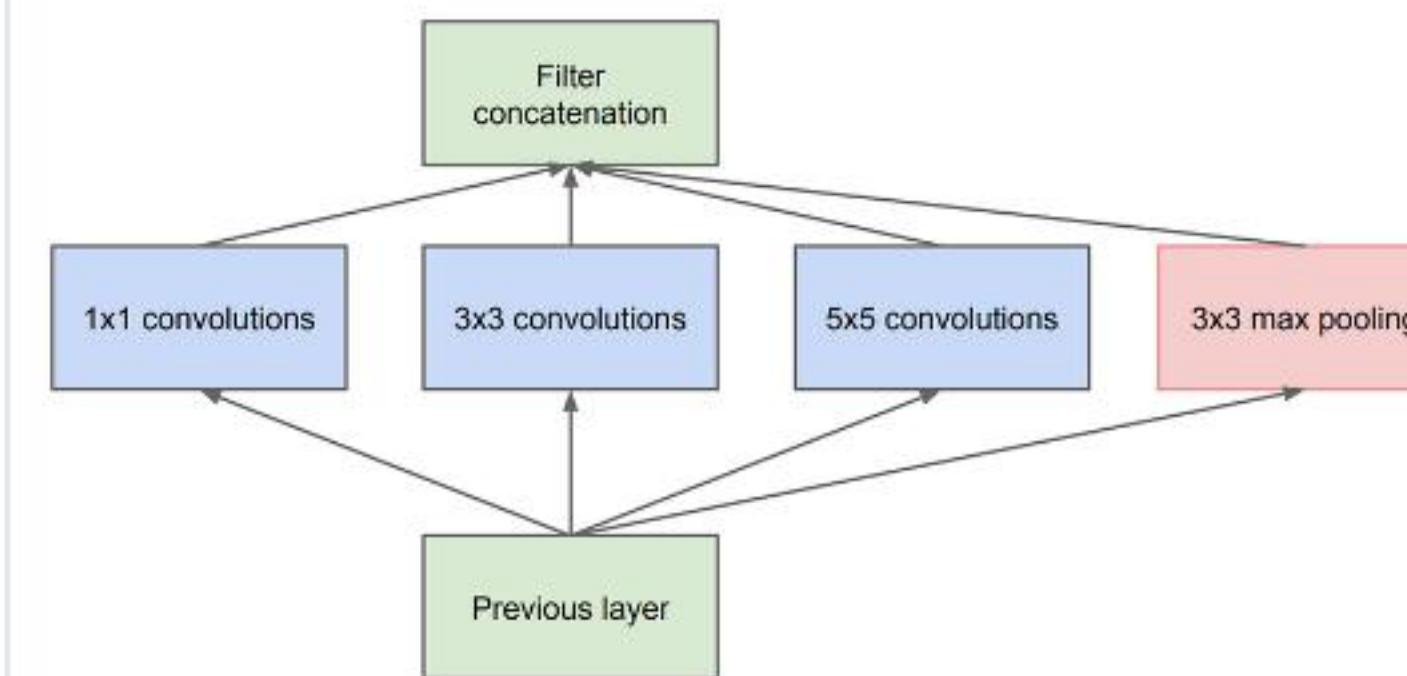
🔗 **Branch 1:** 1×1 Convolution

🔗 **Branch 2:** 3×3 Convolution

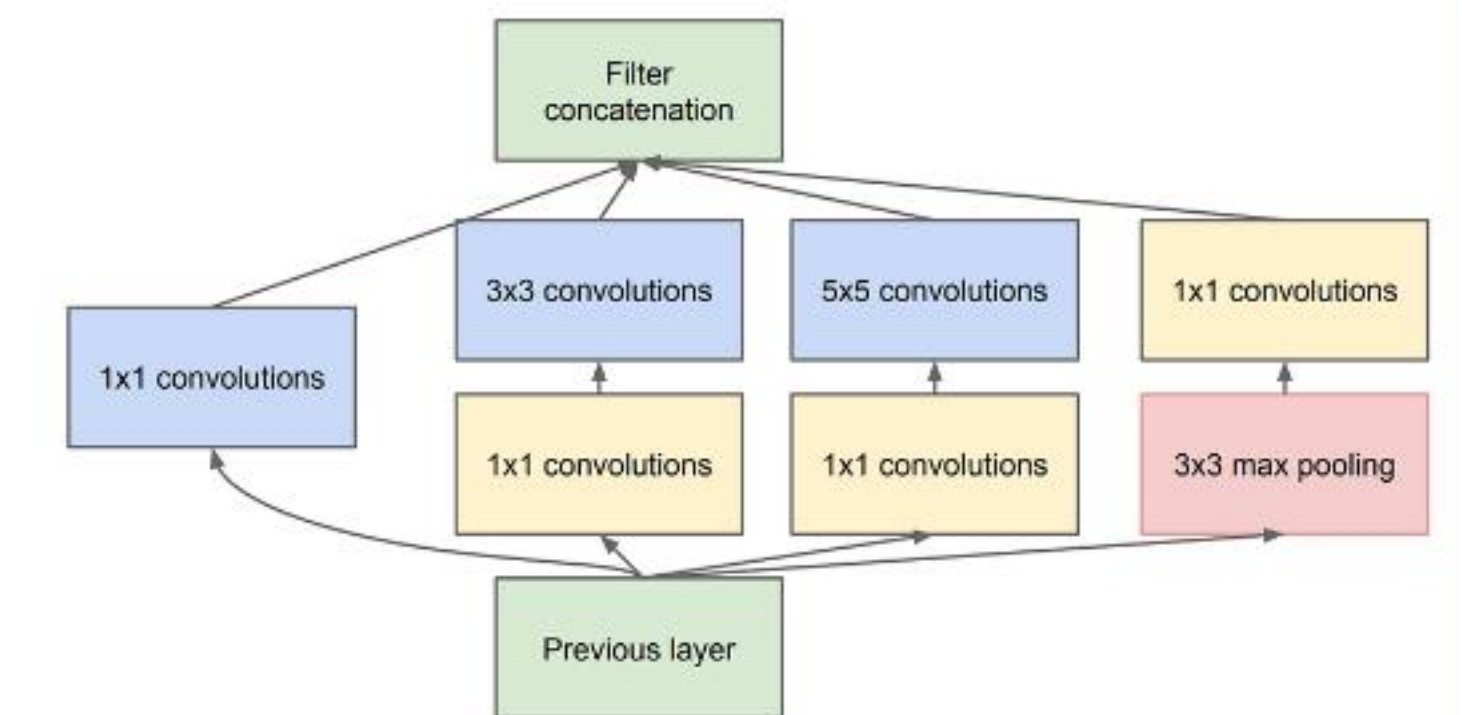
🔗 **Branch 3:** 5×5 Convolution

🔗 **Branch 4:** 3×3 Max Pooling

📦 **Concatenation:** The feature maps from all branches are stacked together along the channel dimension.



(a) Inception module, naïve version



(b) Inception module with dimension reductions

The Computational Problem

The Bottleneck

Running a 5×5 convolution on an input with hundreds of channels (e.g., 512 channels) is computationally disastrous.

$$\text{Computation} \propto (\text{Channels}_{\text{in}} \times \text{Filters} \times 5 \times 5)$$

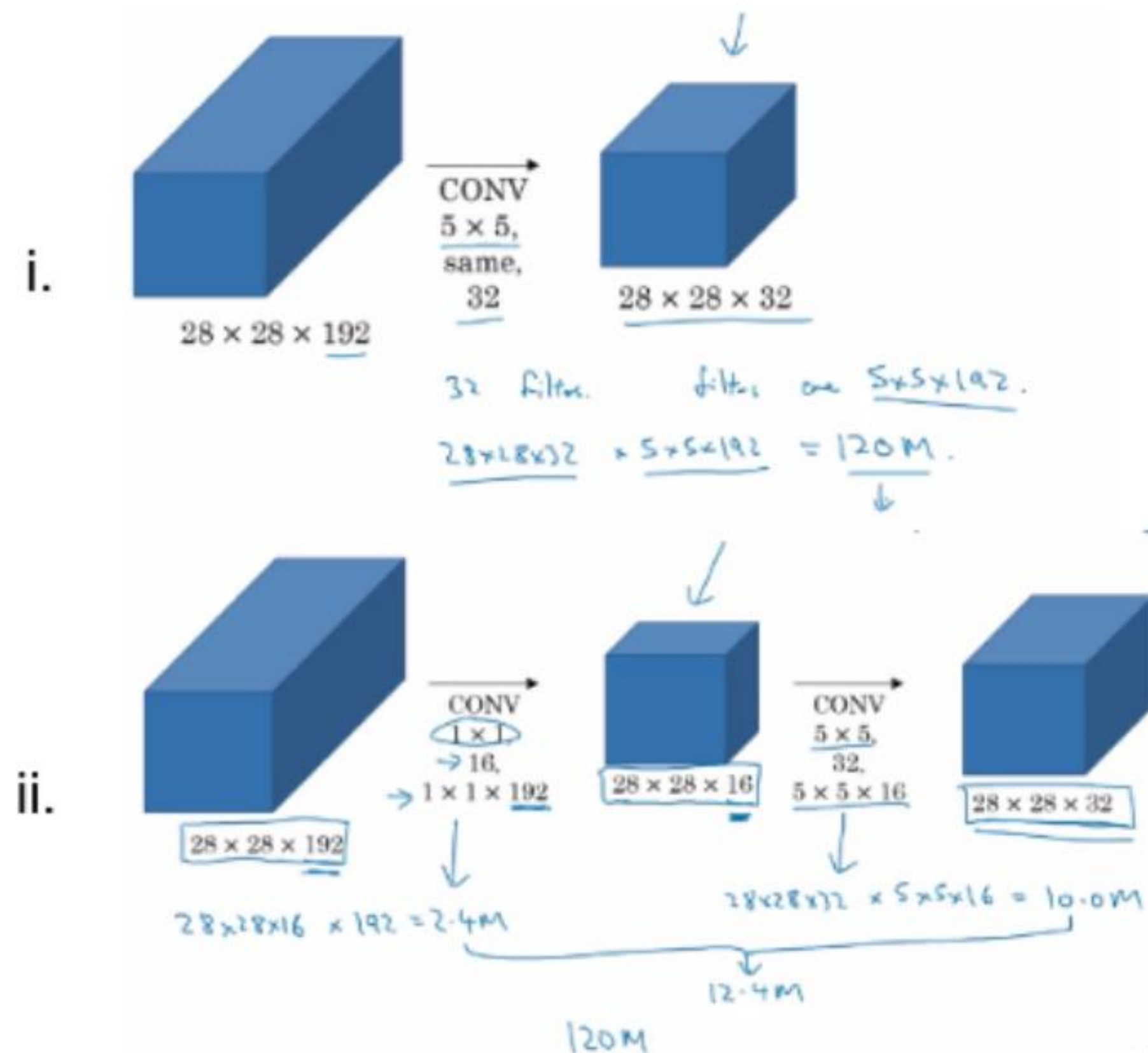
The Question

How can we keep multi-scale parallel processing without an enormous computational cost?

If we concatenate branches, the channel depth explodes layer after layer.

The Power of 1×1 Convolutions

- Also, By using 1×1 filter you can reduce 120M to 12M by a factor of 10.



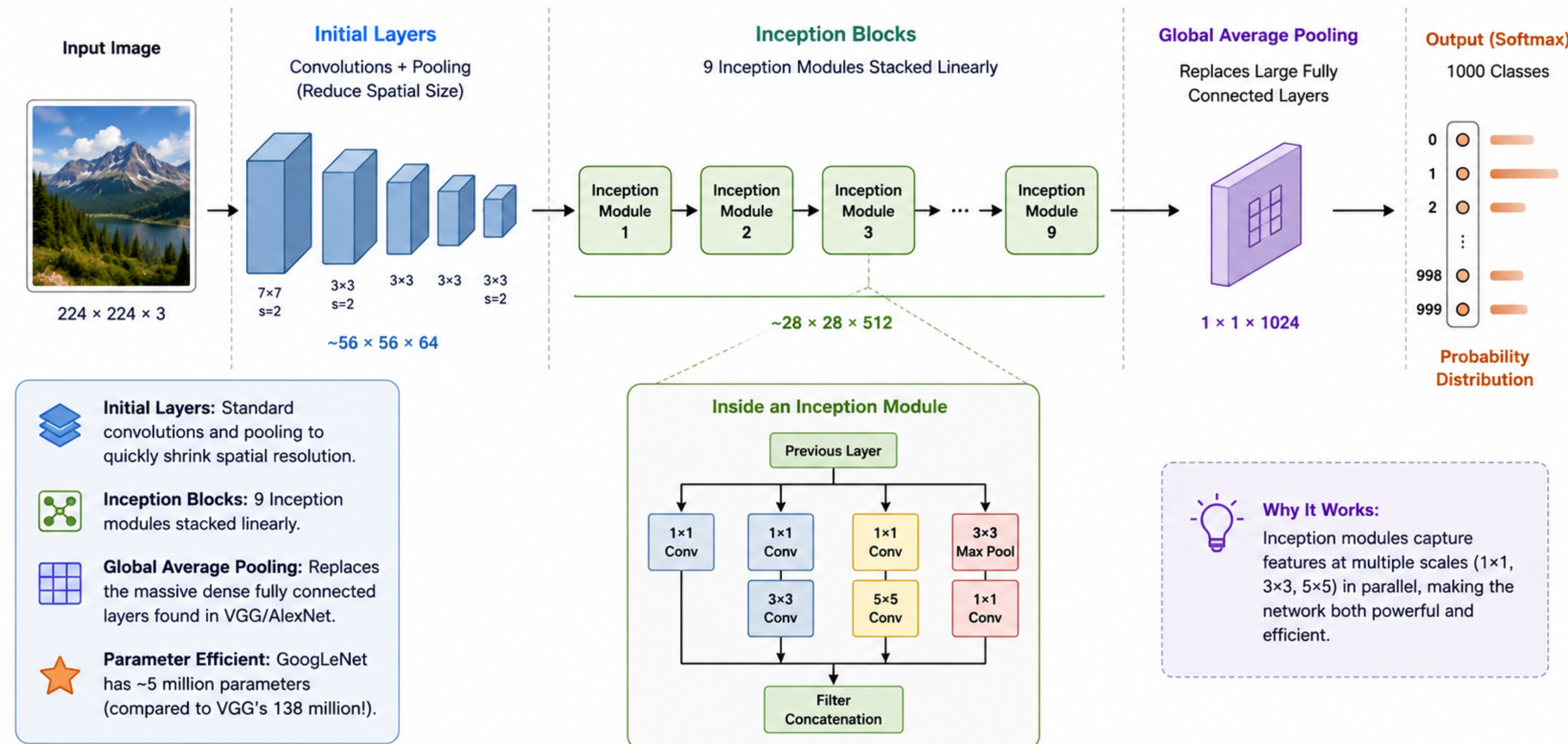
1×1 Convolutions act as **Dimensionality Reduction**.

- ✖ **Shrinks Volume:** Placed *before* the 3×3 and 5×5 convolutions to compress the number of channels.
- ⚙ **Reduces Computation:** Drastically lowers the required multiplication operations for the heavy filters.
- ⌋ **Adds Non-Linearity:** Every 1×1 adds another ReLU activation, making the network smarter.
- ✓ Makes the Inception architecture practically trainable.

Inside GoogLeNet (Inception v1)

- ▶ **Initial Layers:** Standard convolutions and pooling to quickly shrink spatial resolution.
- 📦 **Inception Blocks:** 9 Inception modules stacked linearly.
- ⚡ **Global Average Pooling:** Replaces the massive dense fully connected layers found in VGG/AlexNet.
- ✍️ **Parameter Efficient:** GoogLeNet has ~5 million parameters (compared to VGG's 138 million!).

Inside GoogLeNet (Inception v1)



Evolution of Inception

Inception v2/v3

Batch Normalization &
Factorized Convolutions (e.g.,
5x5 split into two 3x3s).

Inception-ResNet

Combined the parallel
branches of Inception with the
skip connections of ResNet!

Inception v1

(GoogLeNet)
The original multi-scale
architecture with 1x1 dims
reduction.

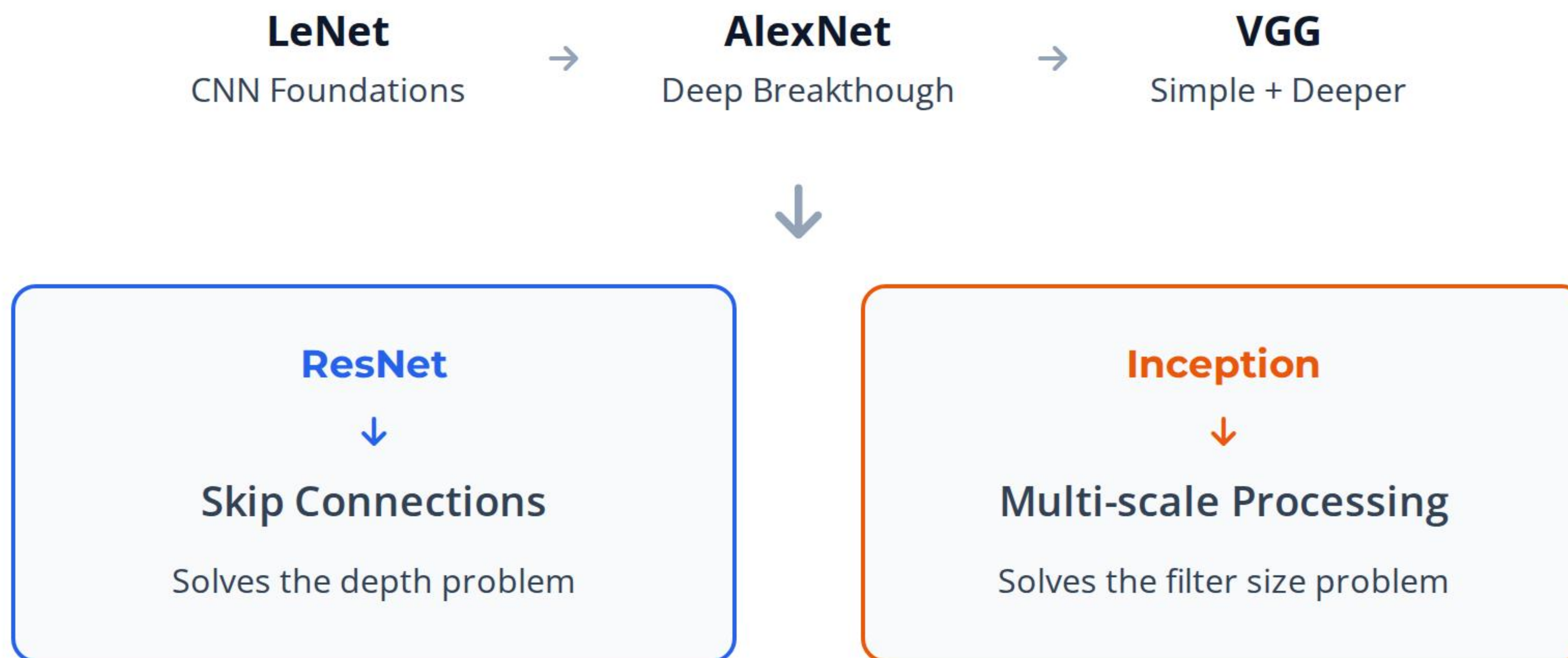
Inception v4

More uniform, simplified, and
deeply optimized architecture.

ResNet vs Inception

Aspect	ResNet	Inception (GoogLeNet)
Introduced	2015 (Microsoft)	2014 (Google)
Core Philosophy	"Go deeper using skip connections."	"Go wider and multi-scale using parallel filters."
Main Architecture	Residual Blocks (addition)	Inception Modules (concatenation)
Computational Strategy	Bottleneck blocks (1×1)	Dimensionality reduction (1×1)
Major Advantage	Allows theoretically infinite depth.	Highly parameter efficient & captures varying scales.

Architecture Evolution: The Bigger Picture



"Modern CNN architectures are not just deeper — they are designed to make information and gradients flow efficiently."

☰ Key Takeaways

- Increasing depth alone can cause degradation (not overfitting).
- ResNet introduces residual learning and skip connections to bypass degradation.
- Inception processes features at multiple spatial scales simultaneously.
- 1×1 convolutions make Inception computationally viable (dimensionality reduction).
- ResNet and Inception represent two distinct solutions to designing better CNNs.

📌 Check Your Understanding

1. Why can a 56-layer CNN perform worse than a 20-layer CNN during training?
2. What does the skip connection actually do in a ResNet block?
3. Why does Inception use 1×1 convolutions before larger filters?

Preview: Week 5 – Regularization

Dropout, Weight Decay, Batch Normalization & Data Augmentation

Image Sources

600 × 400

<http://cv-tricks.com/wp-content/uploads/2017/04/deep-learning-resnet-degradation-problem.png>

Source: cv-tricks.com

600 × 400

https://upload.wikimedia.org/wikipedia/commons/thumb/a/a2/ResNet_residual_block.svg/1200px-ResNet_residual_block.svg.png

Source: en.wikipedia.org

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Source: pub.towardsai.net

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Source: towardsdatascience.com

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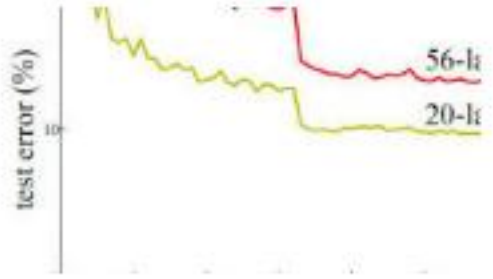
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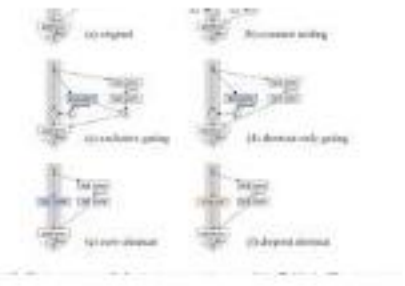
Source: theaisummer.com

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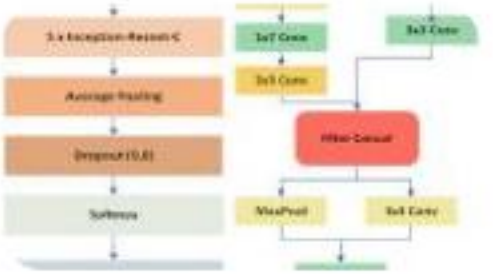
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Source: medium.com



<https://shuzhanfan.github.io/assets/images/2018-11-02/resnet10.jpg>

Source: shuzhanfan.github.io



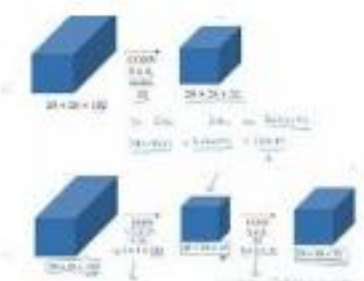
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Source: www.icck.org



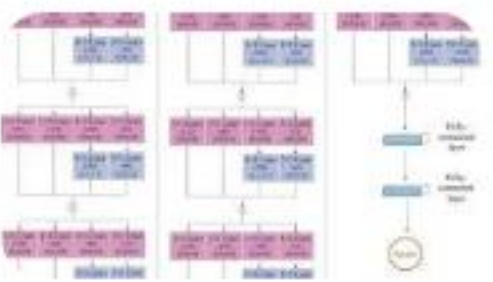
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Source: medium.com



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Source: stats.stackexchange.com



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Source: www.tandfonline.com